

DHILIBAN M
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18. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.

Aim:

To install VMware Workstation and create virtual machines with different operating systems such as Ubuntu Linux on the existing Windows environment.

Procedure:

Step 1: Open VMware Workstation

1. Open VMware Workstation in Windows.
2. Click Create a New Virtual Machine.

Step 2: Select Installation Method

1. Select Typical (recommended).
2. Click Next.

Step 3: Select the Operating System ISO

1. Select Installer disc image file (ISO).
2. Click Browse.
3. Select the Linux/Windows ISO file.
4. Click Next.

Example:

Ubuntu ISO → ubuntu.iso

Step 4: Select Operating System

1. Select the appropriate operating system.
2. For Ubuntu, select:

Linux

Ubuntu 64-bit

3. Click Next.

Step 5: Give Virtual Machine Name

Enter a name such as:

Ubuntu-VM

Select a location where you want to store the VM.

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Click Next.

Step 6: Configure Disk

1. Specify the disk size.

For example:

20 GB

2. Select Store virtual disk as a single file.
3. Click Next.

Step 7: Customize Hardware

Click Customize Hardware.

Configure:

- Memory: 2 GB or more
- Processor: 2 cores
- Network Adapter: NAT

Click Close → Finish.

Step 8: Install the Operating System

1. Start the virtual machine by clicking Power on this virtual machine.
2. The operating system installer will start.
3. Follow the installation instructions.
4. Create a username and password when requested.
5. Wait for the installation to complete.

Step 9: Restart the Virtual Machine

After installation:

1. Restart the VM.
2. Remove the ISO if VMware asks you to do so.
3. Ubuntu will boot from the virtual hard disk.

Step 10: Verify the Installation

Open the Terminal in Ubuntu and run:

```
uname -a
```

Then:

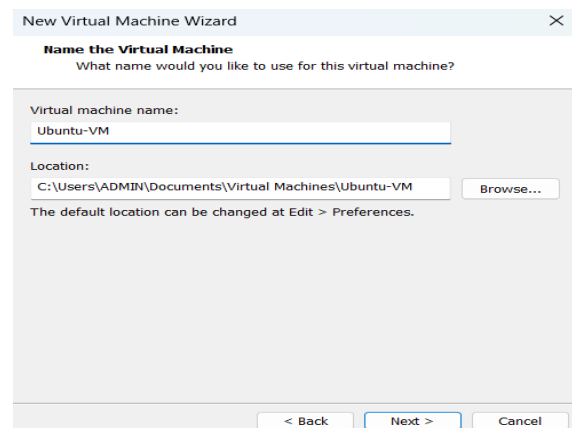
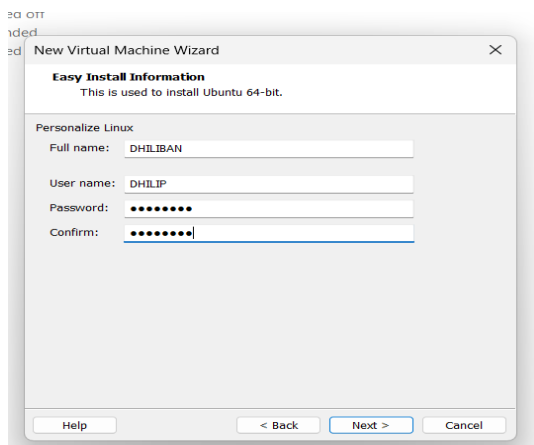
```
ls
```

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And

Pwd

Implementation:



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New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

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Next >

Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

The virtual machine will be created with the following settings:

Name:	Ubuntu-VM
Location:	C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM
Version:	Workstation Beta
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...

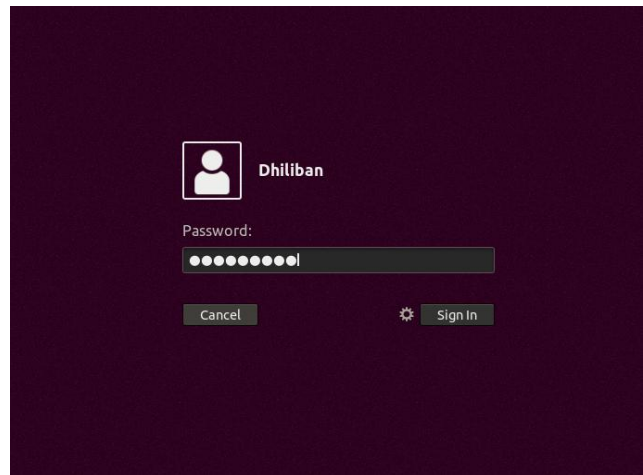
Customize Hardware...

☒ Power on this virtual machine after creation

< Back

Finish

Cancel



```
Activities Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 201
9 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$ pwd
/home/dhilip
dhilip@ubuntu:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
dhilip@ubuntu:~$
```

```
Activities Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 18.04.3 LTS
Release: 18.04
Codename: bionic
dhilip@ubuntu:~$
```

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19. Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.

Aim:

To install a C compiler (GCC) in the Ubuntu virtual machine created using VMware Workstation and execute a simple C program.

Procedure:

Step 1: Start Ubuntu VM

1. Open VMware Workstation.
2. Start your Ubuntu VM.
3. Login to Ubuntu.

Step 2: Open Terminal

Press:

Ctrl + Alt + T

Step 3: Update the Package List

Type:

`sudo apt update`

Press Enter.

Enter your Ubuntu password when asked.

While typing the password, you won't see any characters. This is normal. Type the password and press Enter.

Wait until the update finishes.

Step 4: Install GCC Compiler

Type:

`sudo apt install gcc`

If it asks:

Do you want to continue? [Y/n]

Type:

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Y

and press Enter.

Wait for the installation to complete.

Step 5: Check GCC Installation

Type:

```
gcc --version
```

You should see something similar to:

```
gcc (Ubuntu ...) ...
```

This confirms that the C compiler is installed.

Step 6: Create a C Program

Create a file named hello.c:

```
nano hello.c
```

The Nano editor will open.

Type this program:

```
#include <stdio.h>
```

```
int main()  
{  
    printf("Hello World!\n");  
    return 0;  
}
```

Step 7: Save the Program

In Nano:

1. Press Ctrl + O
2. Press Enter
3. Press Ctrl + X

You will return to the terminal.

Step 8: Compile the C Program

Type:

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gcc hello.c -o hello

If there is no error message, compilation was successful.

Step 9: Execute the Program

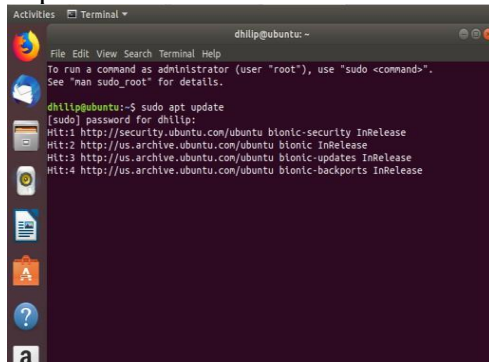
Type:

./hello

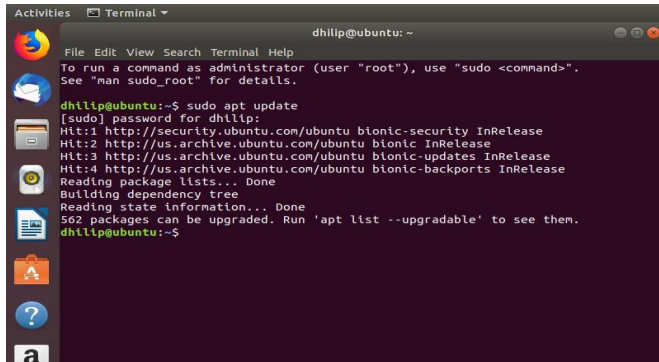
Expected output:

Hello World!

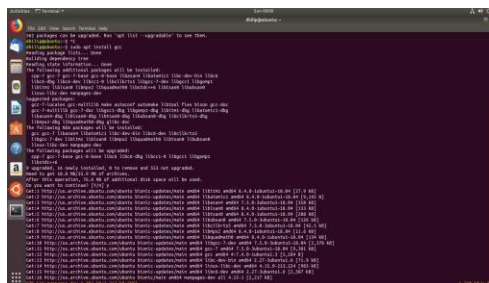
Implementation:



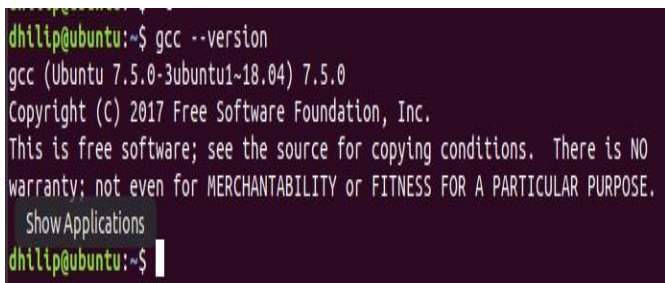
```
dhilip@ubuntu: ~  
File Edit View Search Terminal Help  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
dhilip@ubuntu:~$ sudo apt update  
[sudo] password for dhilip:  
Hit:1 http://security.ubuntu.com/ubuntu bionic-security InRelease  
Hit:2 http://us.archive.ubuntu.com/ubuntu bionic InRelease  
Hit:3 http://us.archive.ubuntu.com/ubuntu bionic-updates InRelease  
Hit:4 http://us.archive.ubuntu.com/ubuntu bionic-backports InRelease
```



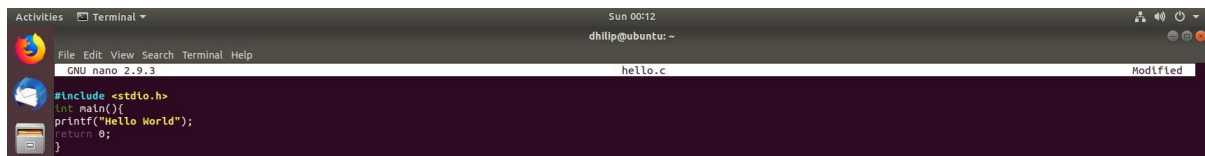
```
dhilip@ubuntu: ~  
File Edit View Search Terminal Help  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
dhilip@ubuntu:~$ sudo apt update  
[sudo] password for dhilip:  
Hit:1 http://security.ubuntu.com/ubuntu bionic-security InRelease  
Hit:2 http://us.archive.ubuntu.com/ubuntu bionic InRelease  
Hit:3 http://us.archive.ubuntu.com/ubuntu bionic-updates InRelease  
Hit:4 http://us.archive.ubuntu.com/ubuntu bionic-backports InRelease  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
562 packages can be upgraded. Run 'apt list --upgradable' to see them.  
dhilip@ubuntu:~$
```



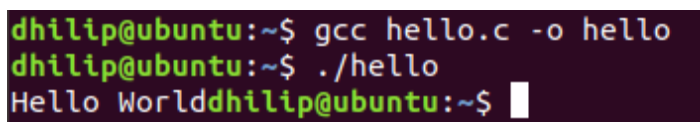
```
dhilip@ubuntu:~$ gcc --version  
gcc (Ubuntu 7.5.0-3ubuntu1~18.04) 7.5.0  
Copyright (c) 2017 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
Show Applications  
dhilip@ubuntu:~$
```



```
dhilip@ubuntu:~$ gcc --version  
gcc (Ubuntu 7.5.0-3ubuntu1~18.04) 7.5.0  
Copyright (c) 2017 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
Show Applications  
dhilip@ubuntu:~$
```



```
Sun 00:12  
dhilip@ubuntu: ~  
GNU nano 2.9.3  
hello.c  
Modified  
#include <stdio.h>  
int main(){  
    printf("Hello World");  
    return 0;  
}
```



```
dhilip@ubuntu:~$ gcc hello.c -o hello  
dhilip@ubuntu:~$ ./hello  
Hello World  
dhilip@ubuntu:~$
```


20. Install Virtualbox with different flavours of linux or windows OS on top of windows 10 using custom installation.

Aim:

To install Oracle VirtualBox on Windows 10 and install a different flavour of Linux/Windows operating system using Custom Installation.

Procedure:

Step 1: Open VirtualBox

1. Open Oracle VM VirtualBox on your Windows computer.
2. Click New to create a new virtual machine.

Step 2: Create the Virtual Machine

1. Enter the VM name, for example:

Ubuntu-VirtualBox

2. Select the location where you want to store the VM.
3. Select the ISO Image of Ubuntu.
4. Select the operating system type:
 - o Type: Linux
 - o Version: Ubuntu (64-bit)
5. Click Next.

Step 3: Select Custom Installation

When VirtualBox provides the installation options, choose the option for custom/unattended installation settings as applicable.

If you see Skip Unattended Installation, select it if you want to perform the OS installation manually.

Click Next.

Step 4: Configure Hardware

Set the hardware resources:

Memory:

2048 MB (2 GB)

Processors:

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2 CPUs

Click Next.

Step 5: Create Virtual Hard Disk

1. Select Create a Virtual Hard Disk Now.
2. Set the disk size to:

20 GB

3. Click Next.
4. Review the configuration.
5. Click Finish.

Step 6: Start the Virtual Machine

1. Select Ubuntu-VirtualBox.
2. Click Start.
3. Ubuntu will boot from the selected ISO file.

Step 7: Install Ubuntu

When the Ubuntu installer appears:

1. Select Install Ubuntu.
2. Select the keyboard layout.
3. Choose the installation type.
4. Select Erase disk and install Ubuntu.

This refers to the virtual hard disk, not your actual Windows C: drive.

5. Set your location/time zone.
6. Create a username and password.
7. Click Install.
8. Wait for the installation to complete.

Step 8: Restart the VM

After installation:

1. Click Restart Now.
2. If VirtualBox asks you to remove the installation medium, press Enter.
3. Ubuntu will restart.
4. Login using the username and password you created.

Step 9: Verify the Installation

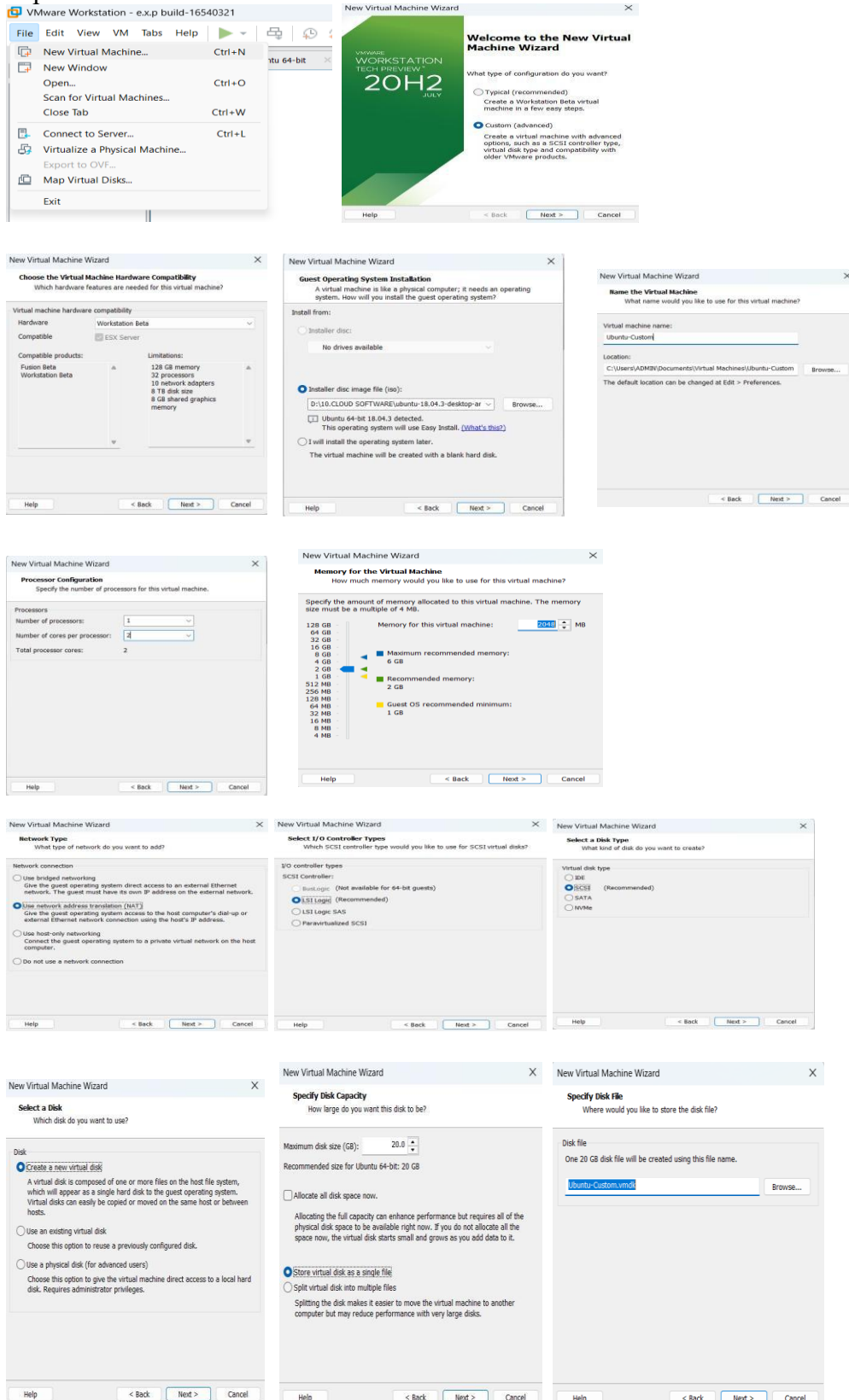
Open Terminal:

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Run:

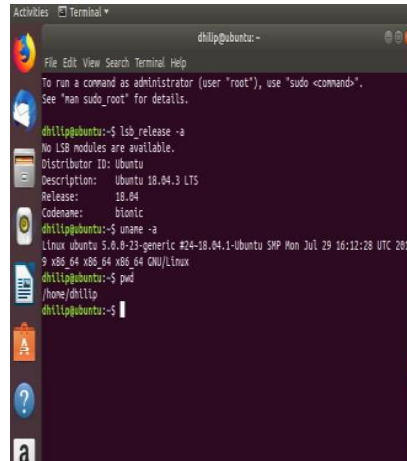
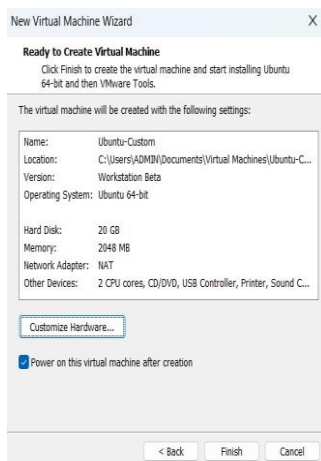
lsb_release -a

Implementation:



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21. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows 7 or 8.

Aim:

To install VMware Workstation and create virtual machines with different operating systems such as Ubuntu Linux on the existing Windows environment.

Procedure:

Step 1: Open VMware Workstation

3. Open VMware Workstation in Windows.
4. Click Create a New Virtual Machine.

Step 2: Select Installation Method

3. Select Typical (recommended).
4. Click Next.

Step 3: Select the Operating System ISO

5. Select Installer disc image file (ISO).
6. Click Browse.
7. Select the Linux/Windows ISO file.
8. Click Next.

Example:

Ubuntu ISO → ubuntu.iso

Step 4: Select Operating System

3. Select the appropriate operating system.
4. For Ubuntu, select:

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Linux
Ubuntu 64-bit

4. Click Next.

Step 5: Give Virtual Machine Name

Enter a name such as:

Ubuntu-VM

Select a location where you want to store the VM.

Click Next.

Step 6: Configure Disk

2. Specify the disk size.

For example:

20 GB

4. Select Store virtual disk as a single file.
5. Click Next.

Step 7: Customize Hardware

Click Customize Hardware.

Configure:

- Memory: 2 GB or more
- Processor: 2 cores
- Network Adapter: NAT

Click Close → Finish.

Step 8: Install the Operating System

6. Start the virtual machine by clicking Power on this virtual machine.
7. The operating system installer will start.
8. Follow the installation instructions.
9. Create a username and password when requested.
10. Wait for the installation to complete.

Step 9: Restart the Virtual Machine

After installation:

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4. Restart the VM.
5. Remove the ISO if VMware asks you to do so.
6. Ubuntu will boot from the virtual hard disk.

Step 10: Verify the Installation

Open the Terminal in Ubuntu and run:

`uname -a`

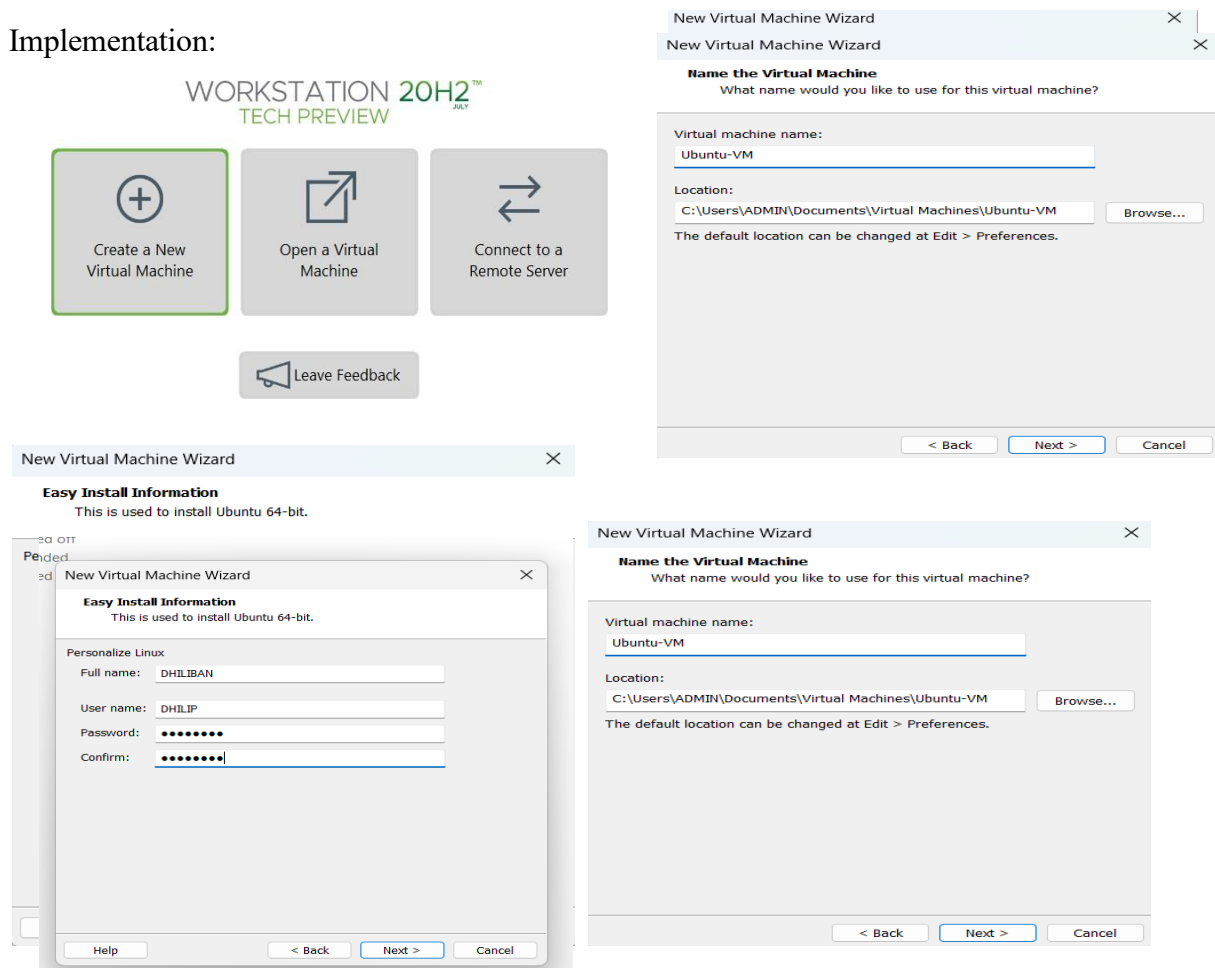
Then:

`ls`

And:

`Pwd`

Implementation:



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New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

The virtual machine will be created with the following settings:

Name:	Ubuntu-VM
Location:	C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM
Version:	Workstation Beta
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...

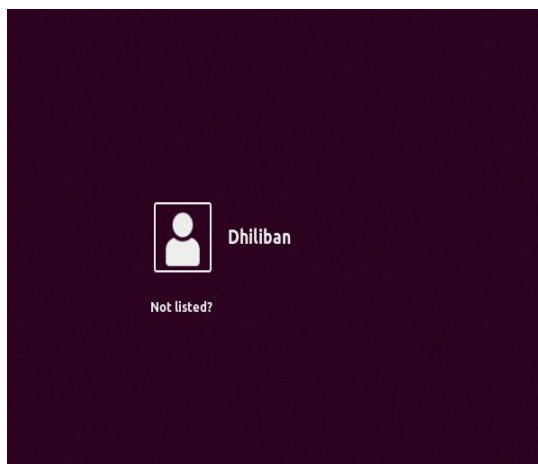
[Customize Hardware...](#)

☒ Power on this virtual machine after creation

< Back

Finish

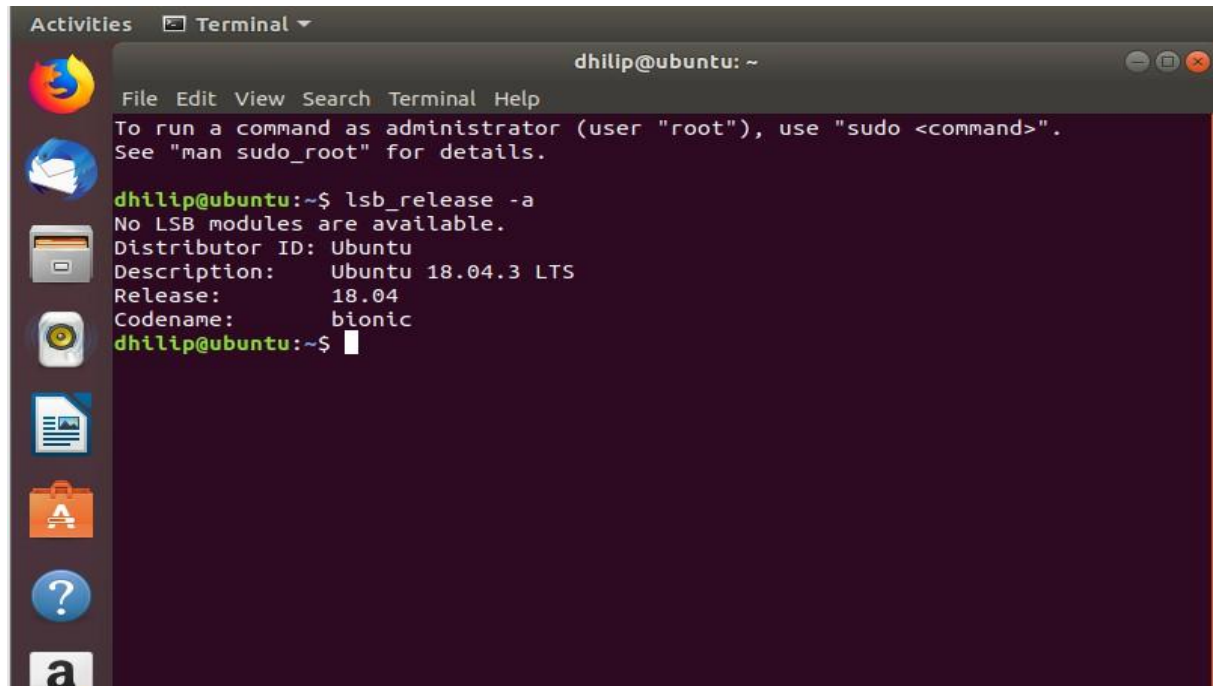
Cancel



```
Activities Terminal
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 201
9 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$ pwd
/home/dhilip
dhilip@ubuntu:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
dhilip@ubuntu:~$
```


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The image shows a terminal window titled "dhilip@ubuntu: ~". The window has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The terminal output shows the command "lsb_release -a" being executed, resulting in the following information: "No LSB modules are available.", "Distributor ID: Ubuntu", "Description: Ubuntu 18.04.3 LTS", "Release: 18.04", and "Codename: bionic". The prompt "dhilip@ubuntu:~\$" is visible at the bottom of the terminal window.

```
dhilip@ubuntu: ~  
File Edit View Search Terminal Help  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
dhilip@ubuntu:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 18.04.3 LTS  
Release:       18.04  
Codename:      bionic  
dhilip@ubuntu:~$
```

22.To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

Aim:

To run virtual machines with different hardware configurations in VMware Workstation and determine how many virtual machines can be utilized simultaneously on the host computer.

Procedure:

Step 1 — Open VMware Workstation

1. Open VMware Workstation Pro.
2. Make sure your previously created Ubuntu VMs are available.

Step 2 — Check the Host Computer Resources

Before starting multiple VMs, check your Windows computer's:

- RAM
- CPU
- Available disk space

You can check RAM and CPU using:

Task Manager → Performance

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Step 3 — Configure the First VM

For example, configure Ubuntu VM 1 with:

RAM: 2 GB
CPU: 1 processor, 2 cores

Start the VM.

Step 4 — Configure the Second VM

Create/select another VM and configure it with:

RAM: 2 GB
CPU: 1 processor, 2 cores

Start the second VM without shutting down the first VM.

Step 5 — Configure Additional VMs

If your computer has sufficient resources, start another VM.

For example:

VM 1 → 2 GB RAM
VM 2 → 2 GB RAM
VM 3 → 2 GB RAM

Do not start more VMs if the host system becomes very slow or runs out of memory.

Step 6 — Check Running Virtual Machines

In VMware Workstation, observe which VMs are powered on.

You can also check Windows:

Task Manager → Performance → Memory / CPU

Observe the resource usage while the VMs are running.

Step 7 — Test Each VM

Open the Terminal in each running Ubuntu VM and execute:

`uname -a`

and:

`pwd`

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If the commands execute successfully, the VMs are running correctly.

Step 8 — Determine the Maximum Number

Start the VMs one by one.

After starting each VM, check:

- CPU usage
- RAM usage
- System responsiveness
- Whether the VM operates normally

The maximum number of VMs is the number that can run simultaneously while the host and VMs remain usable.

Example

If your computer has 8 GB RAM, you might test:

VM 1 → 2 GB

VM 2 → 2 GB

VM 3 → 2 GB

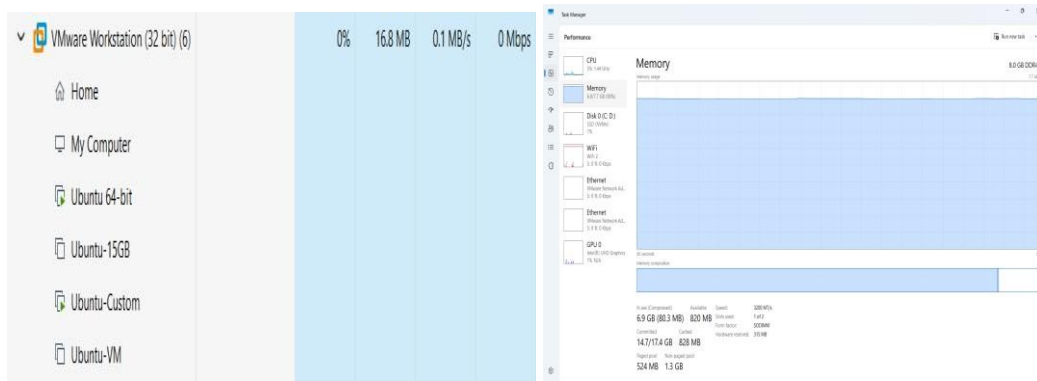
Implementation:

The screenshot displays the VMware Workstation interface. The main window shows the 'Virtual Machine Settings' dialog for 'Ubuntu 64-bit'. The 'Memory' tab is selected, showing a memory allocation of 2048 MB. The 'Summary' tab shows the VM's configuration: 2 processors, 10 GB hard disk, 8 GB hard disk 2, and various controllers. Below the settings, a terminal window is open, showing the output of the 'uname -a' command:

```
dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24-18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 201
9 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$
```

On the right side of the interface, a 'My Computer' window shows a list of VMs and their status:

Name	Status
Ubuntu 64-bit	Powered on
Ubuntu-15GB	Powered off
Ubuntu-Custom	Powered on
Ubuntu-VM	Powered off



23. Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

Aim:

To attach a virtual block (virtual hard disk) to a virtual machine and verify whether the data stored on it remains available even after the virtual machine is released or powered off.

Procedure:

We will use your Ubuntu VM in VMware Workstation Pro.

Step 1 — Power Off the VM

1. Open VMware Workstation Pro.
2. Select your Ubuntu VM.
3. Shut down Ubuntu completely.
4. Make sure the VM shows Powered Off.

Do not suspend it. It should be completely powered off.

Step 2 — Open Virtual Machine Settings

1. Select the Ubuntu VM.
2. Click Edit virtual machine settings.
3. The Virtual Machine Settings window will open.

Step 3 — Add a Virtual Hard Disk

1. Click Add...
2. Select:

Hard Disk

3. Click Next.

Step 4 — Select Disk Type

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Select:

SCSI

Then click Next.

Step 5 — Create a New Virtual Disk

Select:

Create a new virtual disk

Click Next.

Step 6 — Set Disk Size

Enter:

1 GB

You can use 1 GB because we only need a small disk for this experiment.

Select:

Store virtual disk as a single file

Click Next.

Step 7 — Finish Adding the Disk

Leave the disk filename as the default.

Click:

Finish

Then click:

OK

The new virtual hard disk is now attached to your Ubuntu VM.

Step 8 — Start Ubuntu

Power on the Ubuntu VM.

Open Terminal:

Ctrl + Alt + T

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Step 9 — Check the New Virtual Disk

Run:

```
lsblk
```

You should see your original disk and a new disk, for example:

```
sda
├── sda1
└── sda2
sdb
```

The newly added disk will usually appear as sdb.

Do not format sda. That is your Ubuntu system disk.

Step 10 — Create a Partition

If the new disk is /dev/sdb, run:

```
sudo fdisk /dev/sdb
```

Inside fdisk:

Press:

```
n
```

Press Enter.

Then press Enter for the default partition options until the partition is created.

Finally press:

```
w
```

This saves the partition.

Step 11 — Format the Partition

Run:

```
sudo mkfs.ext4 /dev/sdb1
```

This creates an ext4 filesystem on the new virtual disk.

Step 12 — Create a Mount Directory

Run:

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```
sudo mkdir /mnt/virtualdisk
```

Step 13 — Mount the Virtual Disk

Run:

```
sudo mount /dev/sdb1 /mnt/virtualdisk
```

Check:

```
df -h
```

You should see /mnt/virtualdisk.

Step 14 — Create Data on the Virtual Block

Create a file:

```
sudo touch /mnt/virtualdisk/test_data.txt
```

Put some text into it:

```
echo "Virtual Block Test Data" | sudo tee /mnt/virtualdisk/test_data.txt
```

Check it:

```
cat /mnt/virtualdisk/test_data.txt
```

Expected:

Virtual Block Test Data

Data has now been stored on the additional virtual disk.

Step 15 — Release/Power Off the VM

1. Shut down Ubuntu:

```
sudo shutdown -h now
```

2. Wait until VMware shows the VM as Powered Off.

Step 16 — Start the VM Again

Power on the same Ubuntu VM.

Open Terminal.

Check the disk:

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lsblk

You should see the additional disk again.

Step 17 — Mount the Virtual Disk Again

Run:

```
sudo mount /dev/sdb1 /mnt/virtualdisk
```

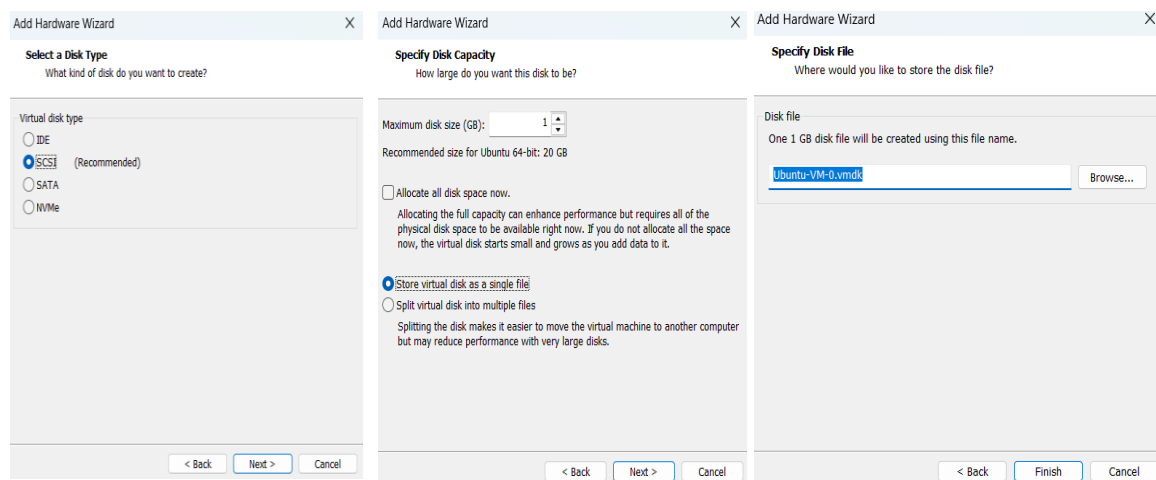
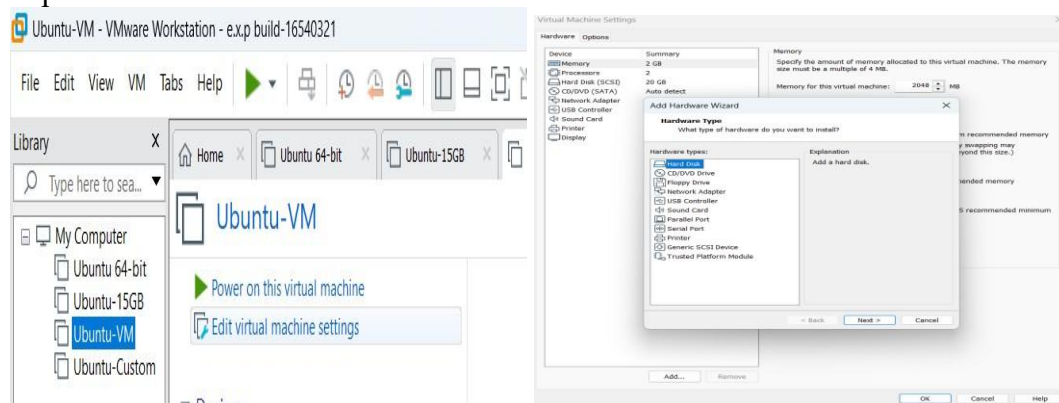
Then:

```
cat /mnt/virtualdisk/test_data.txt
```

Expected Output

Virtual Block Test Data

Implementation:



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▼ Devices

Memory	2 GB
Processors	2
Hard Disk (SCSI)	20 GB
Hard Disk 2 (SCSI)	1 GB
CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

▼ Description

Type here to enter a description of this virtual machine.

Activities Terminal

dhilip@ubuntu: ~

File Edit View Search Terminal Help

```
dhilip@ubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0       7:0      0 1008K  1 loop /snap/gnome-logs/61
loop1       7:1      0  54.4M  1 loop /snap/core18/1066
loop2       7:2      0  14.8M  1 loop /snap/gnome-characters/296
loop3       7:3      0   3.7M  1 loop /snap/gnome-system-monitor/100
loop4       7:4      0  88.5M  1 loop /snap/core/7270
loop5       7:5      0 149.9M  1 loop /snap/gnome-3-28-1804/67
loop6       7:6      0    4M   1 loop /snap/gnome-calculator/406
loop7       7:7      0  42.8M  1 loop /snap/gtk-common-themes/1313
loop8       7:8      0 105.2M  1 loop /snap/core/17292
sda          8:0      0   20G   0 disk
└─sda1       8:1      0   20G   0 part /
sdb          8:16     0    1G   0 disk
sr0         11:0     1 1024M   0 rom

dhilip@ubuntu:~$
```

```
dhilip@ubuntu:~$ sudo fdisk /dev/sdb
[sudo] password for dhilip:

Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x73706a80.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-2097151, default 2048):
Last sector, +sectors or +size[K,M,G,T,P] (2048-2097151, default 2097151):

Created a new partition 1 of type 'Linux' and of size 1023 MiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

dhilip@ubuntu:~$
```

```
dhilip@ubuntu:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.44.1 (24-Mar-2018)
Creating filesystem with 261888 4k blocks and 65536 inodes
Filesystem UUID: 9bdc26bb-5eaf-4499-9189-943da8ea1634
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

dhilip@ubuntu:~$
```

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```
dhilip@ubuntu:~$ sudo mkdir /mnt/virtualdisk
dhilip@ubuntu:~$ sudo mount /dev/sdb1 /mnt/virtualdisk
dhilip@ubuntu:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	960M	0	960M	0%	/dev
tmpfs	197M	1.8M	195M	1%	/run
/dev/sda1	20G	7.1G	12G	39%	/
tmpfs	984M	0	984M	0%	/dev/shm
tmpfs	5.0M	4.0K	5.0M	1%	/run/lock
tmpfs	984M	0	984M	0%	/sys/fs/cgroup
/dev/loop0	1.0M	1.0M	0	100%	/snap/gnome-logs/61
/dev/loop1	55M	55M	0	100%	/snap/core18/1066
/dev/loop2	15M	15M	0	100%	/snap/gnome-characters/296
/dev/loop3	3.8M	3.8M	0	100%	/snap/gnome-system-monitor/100
/dev/loop4	89M	89M	0	100%	/snap/core/7270
/dev/loop5	150M	150M	0	100%	/snap/gnome-3-28-1804/67
/dev/loop6	4.2M	4.2M	0	100%	/snap/gnome-calculator/406
/dev/loop7	43M	43M	0	100%	/snap/gtk-common-themes/1313
tmpfs	197M	16K	197M	1%	/run/user/121
tmpfs	197M	24K	197M	1%	/run/user/1000
/dev/loop8	106M	106M	0	100%	/snap/core/17292
/dev/loop9	67M	67M	0	100%	/snap/core24/1643
/dev/loop10	128K	128K	0	100%	/snap/bare/5
/dev/loop11	896K	896K	0	100%	/snap/gnome-logs/129
/dev/loop12	1.8M	1.8M	0	100%	/snap/gnome-system-monitor/193
/dev/loop13	9.5M	9.5M	0	100%	/snap/gnome-characters/805
/dev/loop14	2.2M	2.2M	0	100%	/snap/gnome-calculator/972
/dev/loop15	92M	92M	0	100%	/snap/gtk-common-themes/1535
/dev/loop16	165M	165M	0	100%	/snap/gnome-3-28-1804/198
/dev/loop17	402M	402M	0	100%	/snap/mesa-2404/1839
/dev/sdb1	991M	2.6M	922M	1%	/mnt/virtualdisk

```
dhilip@ubuntu:~$
```

```
dhilip@ubuntu:~$ echo "Virtual Block Test Data" | sudo tee /mnt/virtualdisk/test_data.txt
Virtual Block Test Data
```

```
dhilip@ubuntu:~$ cat /mnt/virtualdisk/test_data.txt
Virtual Block Test Data
dhilip@ubuntu:~$
```

Activities Terminal Sun 02:13 dhilip@ubuntu: ~

```
dhilip@ubuntu:~$ lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
loop0	7:0	0	1.7M	1	loop	/snap/gnome-system-monitor/193
loop1	7:1	0	91.7M	1	loop	/snap/gtk-common-themes/1535
loop2	7:2	0	4M	1	loop	/snap/gnome-calculator/406
loop3	7:3	0	2.1M	1	loop	/snap/gnome-calculator/972
loop4	7:4	0	149.9M	1	loop	/snap/gnome-3-28-1804/67
loop5	7:5	0	105.2M	1	loop	/snap/core/17292
loop6	7:6	0	54.4M	1	loop	/snap/core18/1066
loop7	7:7	0	3.7M	1	loop	/snap/gnome-system-monitor/100
loop8	7:8	0	66.8M	1	loop	/snap/core24/1643
loop9	7:9	0	14.8M	1	loop	/snap/gnome-characters/296
loop10	7:10	0	614.5M	1	loop	/snap/gnome-46-2404/164
loop11	7:11	0	402M	1	loop	/snap/mesa-2404/1839
loop12	7:12	0	164.8M	1	loop	/snap/gnome-3-28-1804/198
loop13	7:13	0	1008K	1	loop	/snap/gnome-logs/61
loop14	7:14	0	9.4M	1	loop	/snap/gnome-characters/805
loop15	7:15	0	4K	1	loop	/snap/bare/5
loop16	7:16	0	42.8M	1	loop	/snap/gtk-common-themes/1313
loop17	7:17	0	860K	1	loop	/snap/gnome-logs/129
sda	8:0	0	20G	0	disk	
└─sda1	8:1	0	20G	0	part	/
sdb	8:16	0	1G	0	disk	
└─sdb1	8:17	0	1023M	0	part	
sr0	11:0	1	1024M	0	rom	

```
dhilip@ubuntu:~$
```

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```
dhilip@ubuntu:~$ sudo mount /dev/sdb1 /mnt/virtualdisk
[sudo] password for dhilip:
dhilip@ubuntu:~$ cat /mnt/virtualdisk/test__data.txt
Virtual Block Test Data
dhilip@ubuntu:~$
```

24.To showcase the virtual machine migration based on the certain condition from one host to the other.

Aim:

To demonstrate the migration of a virtual machine from one host/storage location to another using VMware Workstation Pro and verify the migrated virtual machine by executing basic Linux commands.

Procedure:

1. Power off the Ubuntu VM properly from Ubuntu.
2. Close VMware Workstation Pro.
3. Open File Explorer and navigate to:

C:\Users\ADMIN\Documents\Virtual Machines

4. Locate the original VM folder:

Ubuntu-VM

5. Copy the complete Ubuntu-VM folder.
6. Paste it in the same location and rename the copied folder as:

Ubuntu-VM-Migrated

7. Open VMware Workstation Pro.
8. Select:
File → Open
9. Navigate to:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM-Migrated

10. Select the file with type:
VMware virtual machine configuration (.vmx).
11. Click Open.
12. VMware displays:
—Did you move or copy this virtual machine?||
13. Select:
I Copied It
14. Click OK.

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15. Power on the migrated Ubuntu virtual machine.

16. After Ubuntu starts, open the Terminal using:

Ctrl + Alt + T

17. Verify the migrated VM using:

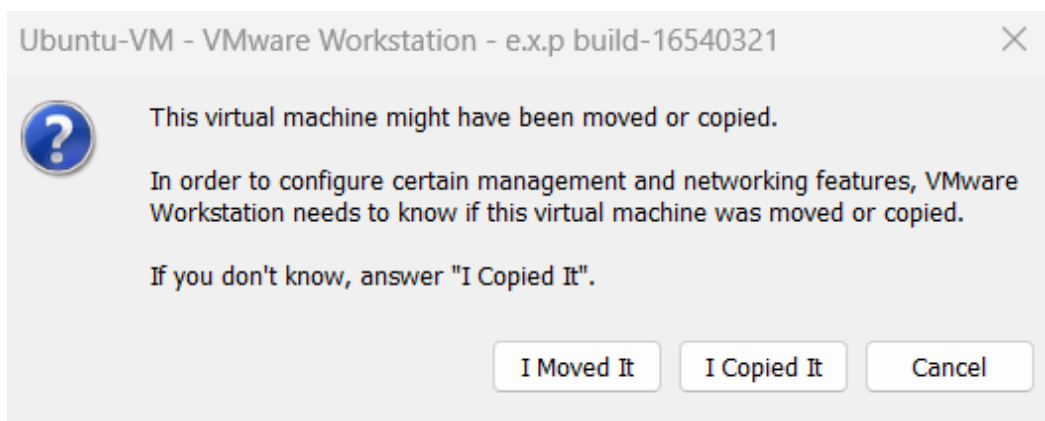
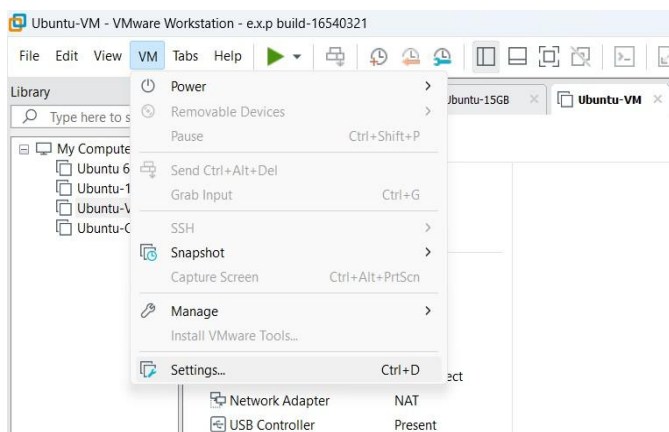
hostname

18. Check the Linux system information using:

uname -a

19. If the Ubuntu VM starts successfully and the commands execute correctly, the migration is successfully demonstrated.

Implementation:



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